

Themis

An Integrated Digital Platform for Legal Aid, Document Automation, and Case Management in India

Functional Requirements, User Stories, Modules, Database Schema & UI Design

Team

A R Devadathan

Abiya John

Amal K R

Abstract

Access to justice in the Indian legal ecosystem remains a significant challenge due to systemic backlogs, complex procedural barriers, and information asymmetry. This paper presents Themis, a comprehensive, production-ready legal tech platform designed to bridge the gap between citizens, legal professionals, and legal aid services. Architected using a high-performance FastAPI backend and a responsive Next.js frontend, the platform integrates relational data management via PostgreSQL with asynchronous task queues powered by Celery and RabbitMQ.

Themis introduces a modular, multi-phase framework that addresses critical legal workflows: advanced legal knowledge search, guided legal assessments, and automated generation of editable complaints and Right to Information (RTI) applications. To ensure secure record-keeping, the system implements an S3-compatible document repository utilizing MinIO, featuring cryptographically signed uploads, automated optical character recognition (OCR) pipelines, and rigorous audit logs. Furthermore, the platform incorporates a verification engine to match eligible citizens with verified legal aid attorneys, alongside an interactive timeline module for automated case and hearing management. By consolidating document orchestration, AI-driven assessments, and secure collaboration into a single, hardened infrastructure, Themis provides a scalable blueprint for modernizing public legal aid delivery and enhancing digital legal literacy.

Stack: FastAPI, Next.js, OCR, Indian Legal System

1. Requirement Collection

Requirements for Themis were gathered by studying the practical barriers citizens face when accessing legal aid in India, including procedural complexity, lack of legal literacy, delayed grievance redressal, and difficulty locating verified legal aid attorneys. The consolidated functional and non-functional requirements below define the scope of the platform.

1.1 Functional Requirements

1.1.1 Legal Knowledge Search

Users should be able to search a structured legal knowledge base covering statutes, procedures, and citizen rights using natural-language queries, returning simplified, understandable explanations rather than raw legal text.

1.1.2 Guided Legal Assessment

The platform should offer a guided, question-based assessment flow that helps users identify the nature of their legal issue and determines the most relevant next steps, forms, or authorities to approach.

1.1.3 Automated Document Generation

Themis should automatically generate editable legal documents such as complaints and Right to Information (RTI) applications, pre-filled using the user's assessment responses and personal details.

1.1.4 Secure Document Repository

The system should provide an S3-compatible document storage layer (MinIO) supporting cryptographically signed uploads, versioning, automated OCR processing of scanned documents, and complete audit trails of access and modification.

1.1.5 Legal Aid Attorney Matching

A verification engine should match eligible citizens with verified legal aid attorneys based on case type, jurisdiction, and attorney availability, ensuring only credentialed professionals are surfaced to users.

1.1.6 Case and Hearing Timeline Management

An interactive timeline module should track case milestones, hearing dates, and pending actions, sending reminders to both citizens and assigned attorneys as key dates approach.

1.1.7 User Authentication and Role-Based Access

The system should support secure authentication and role-based authorization, with distinct roles such as Citizen, Attorney, Legal Aid Coordinator, and Administrator, each with different permissions.

1.1.8 Asynchronous Processing

Long-running tasks such as OCR extraction, document generation, and notification dispatch should be handled asynchronously via Celery and RabbitMQ so the platform remains responsive under load.

1.2 Non-Functional Requirements

1.2.1 Security

All uploaded documents must be encrypted at rest and in transit, with cryptographically signed upload URLs and strict access control enforced at the storage layer.

1.2.2 Scalability

The platform should support concurrent use by citizens, attorneys, and coordinators across multiple jurisdictions, with background workers scaling independently of the API layer.

1.2.3 Reliability

Document generation and OCR pipelines should be fault-tolerant, with failed background jobs automatically retried via Celery's task queue mechanisms.

1.2.4 Auditability

Every document access, upload, and modification must be logged in an immutable audit trail to support legal accountability and dispute resolution.

1.2.5 Usability

The interface should be accessible to citizens with limited legal or technical literacy, using plain language, guided flows, and minimal jargon throughout.

2. Create Initial User Stories

Citizen: As a citizen, I want to search the legal knowledge base in plain language so that I can understand my rights without needing to consult a lawyer first.

Citizen: As a citizen, I want a guided assessment that asks me simple questions so that I can identify the correct legal action to take for my situation.

Citizen: As a citizen, I want the platform to automatically generate a complaint or RTI application from my answers so that I don't have to draft legal documents myself.

Citizen: As a citizen, I want to securely upload supporting documents so that my case file is complete and accessible whenever I need it.

Citizen: As a citizen, I want to be matched with a verified legal aid attorney so that I can get professional help I can trust.

Attorney: As an attorney, I want to view case files and uploaded documents assigned to me so that I can prepare properly before hearings.

Attorney: As an attorney, I want to update case status and hearing dates so that my clients stay informed about their case progress.

Legal Aid Coordinator: As a coordinator, I want to review attorney verification requests so that only credentialed professionals are onboarded onto the platform.

Legal Aid Coordinator: As a coordinator, I want to monitor case load distribution across attorneys so that legal aid resources are allocated fairly.

Administrator: As an administrator, I want complete audit logs of document access so that I can investigate any compliance or security concerns.

User: As a user, I want to receive reminders about upcoming hearings and deadlines so that I never miss an important case milestone.

3. Identify Modules and Assign Ownership

The platform is divided into three core modules, distributed across the team based on individual focus areas. Shared components are jointly owned to ensure system-wide consistency.

Module	Team Member	Responsibilities
Frontend Dashboard	Abiya John	Design citizen and attorney-facing interfaces, guided assessment flow, document upload UI, case timeline views, responsive layouts, and state management.
Backend API	A R Devadathan	Develop FastAPI services, authentication, REST APIs, database schema, Alembic migrations, Celery task integration, and business logic.
Document, OCR & Case Management Engine	Amal K R	Implement MinIO document repository, signed uploads, OCR processing pipeline, RTI/complaint document

Module	Team Member	Responsibilities
		generation, attorney-matching engine, and case/hearing timeline automation.

Shared Responsibilities

The following components are shared among all three team members because they impact the entire system:

- Docker Compose configuration
- CI/CD pipeline
- System integration
- Database migrations review (Alembic)
- Testing and deployment

4. Create Initial Database Schema and UI

4.1 Initial Database Schema

The schema below models the core entities required to support legal assessments, document generation, secure storage, attorney matching, and case timeline management.

Users

Stores citizen, attorney, coordinator, and administrator account information and roles.

```

id
name
email
password_hash
role
created_at

```

Cases

Represents a citizen's legal matter tracked through the platform.

```

id
citizen_id
attorney_id
case_type
status
jurisdiction
created_at

```

Assessments

Stores guided assessment responses used to determine legal pathways.

```
id
user_id
case_id
responses_json
recommended_action
created_at
```

Documents

Metadata for files stored in the MinIO document repository.

```
id
case_id
uploaded_by
file_key
file_type
ocr_status
uploaded_at
```

Generated Documents

Stores auto-generated complaints and RTI applications.

```
id
case_id
document_type
content
status
generated_at
```

Attorney Verifications

Tracks attorney credential verification status.

```
id
attorney_id
bar_council_id
verification_status
verified_at
```

Hearings

Records scheduled hearing and case milestone events.

```
id
case_id
hearing_date
location
status
```

notes

Audit Logs

Immutable record of document access and modification events.

```
id
user_id
document_id
action
timestamp
```

4.2 Frontend UI Design Plan

Design Philosophy

The Themis interface is designed to feel approachable and trustworthy rather than intimidating, since many users will have limited legal or technical literacy. The design favors plain language, generous whitespace, and a calm visual tone appropriate for a civic legal-aid service, while giving attorneys and coordinators a more information-dense, professional case-management view.

Color Palette

The interface uses a restrained, trust-oriented palette.

- Deep Navy (#1B2A41) — Primary color for headers, navigation, and key actions, conveying institutional trust.
- Warm White (#FAF9F6) — Primary background for high readability and an approachable feel.
- Justice Gold (#C9A24B) — Accent color for highlights, active states, and the Themis emblem.
- Slate Grey (#5B6570) — Secondary text and UI chrome.
- Alert Terracotta (#B5502D) — Reserved for deadlines, overdue actions, and critical case alerts.

Typography

- Source Serif 4 — Used for headings and the Themis wordmark, lending a formal, legal-document feel.
- Inter — Used for body text, forms, and case data, prioritizing clarity and legibility.

Layout

The citizen-facing dashboard follows a simple three-panel layout:

- Left Panel — Navigation between Legal Search, Guided Assessment, My Documents, and My Case.
- Center Panel — Primary workspace: assessment questions, generated document previews, or case timeline.
- Right Panel — Case summary, assigned attorney details, and upcoming hearing reminders.

Signature Visual Element

The dashboard's signature element is the Case Journey Timeline — a vertical, milestone-based progress tracker showing each stage of a citizen's case (Assessment → Document Filed → Attorney Assigned → Hearing

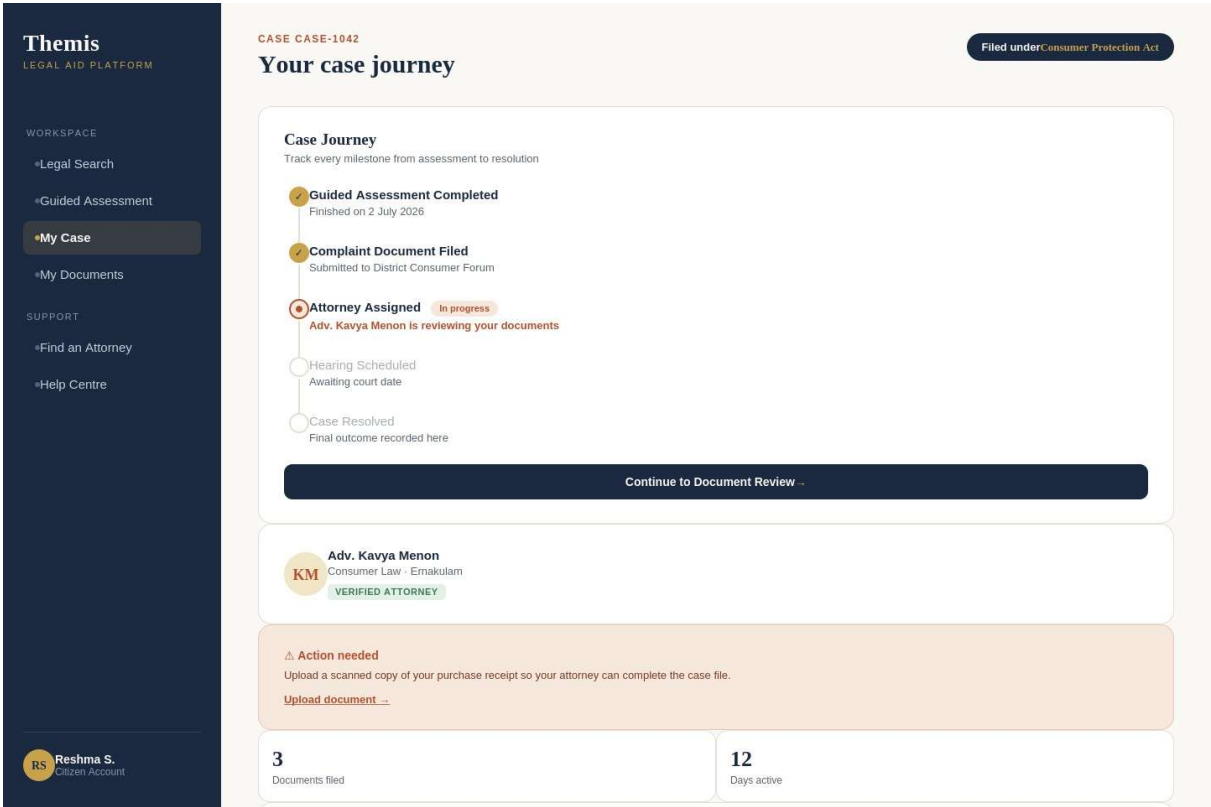
Scheduled → Resolved). Completed stages are marked in Justice Gold, the active stage pulses gently, and upcoming hearing dates are highlighted in Alert Terracotta when approaching, giving users an immediate, reassuring sense of progress through an otherwise opaque legal process.

5. UI Design

The high-fidelity mockups below apply the design tokens defined in the UI Design Plan — Deep Navy, Warm White, Justice Gold, and Alert Terracotta — across the platform's three primary experiences: the citizen dashboard, the guided legal assessment, and the attorney case-management view.

5.1 Citizen Dashboard — Case Journey

The citizen's home screen centers on the Case Journey Timeline, the platform's signature visual element. Completed milestones are marked in Justice Gold, the active stage is highlighted in Alert Terracotta, and a persistent reminder card nudges the citizen toward the next required action.



Citizen Dashboard

Case timeline, assigned attorney, and document status at a glance.

5.2 Guided Legal Assessment

The guided assessment uses a two-panel layout: a persistent progress rail on Deep Navy shows how far along the citizen is, while the right panel presents one plain-language question at a time with icon-led answer cards, minimizing cognitive load for first-time users.

Themis.

GUIDED ASSESSMENT

Let's understand your situation

Answer a few plain-language questions. We'll use your responses to recommend the right legal pathway and pre-fill your documents.

✓ Tell us what happened

2 Type of legal issue

3 Your details

4 Recommended action

QUESTION 2 OF 4

Which best describes the issue you're facing?

This helps Themis recommend the correct complaint format and match you with an attorney who specialises in this area.

☒  **Faulty product or unfair service**
Consumer Protection Act complaint

☐  **Dispute with a landlord or tenant**
Tenancy and rental disputes

☐  **Government department not responding**
Right to Information (RTI) application

☐  **Workplace or employment issue**
Labour grievance

-- Back

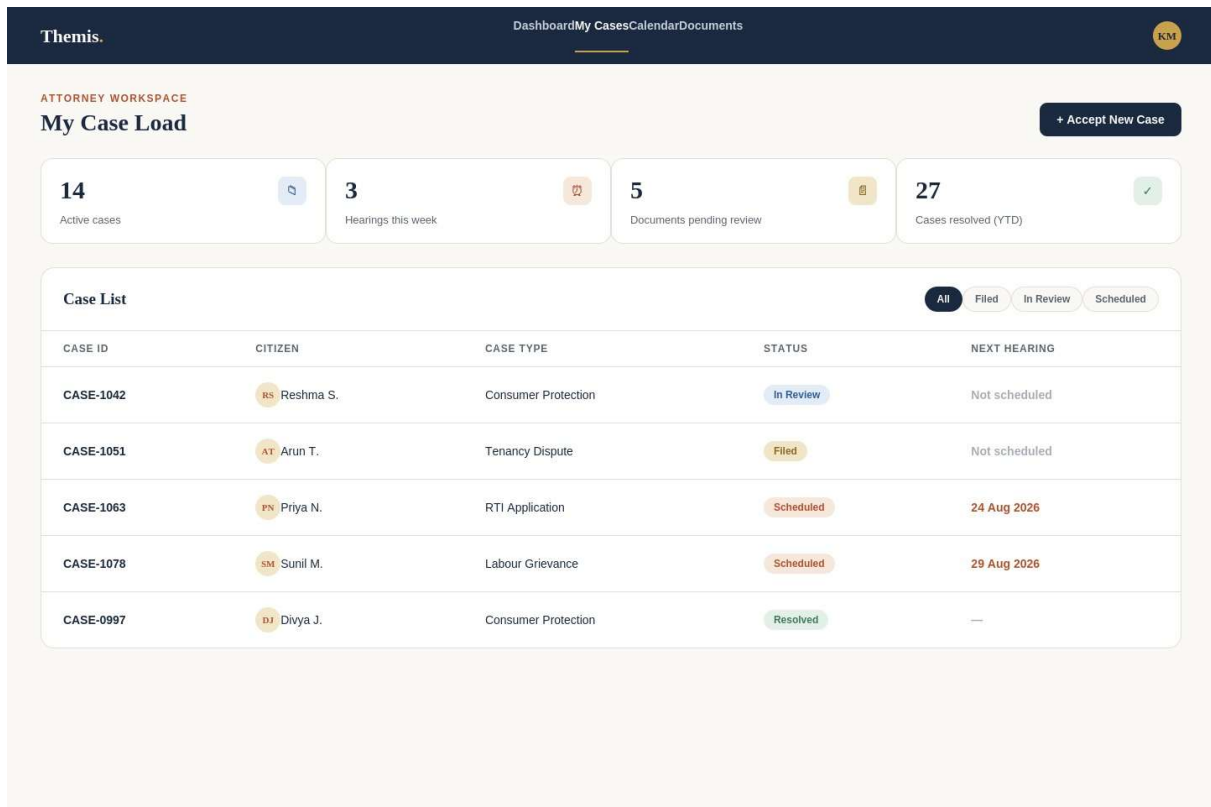
Continue --

Guided Assessment Flow

Step-by-step legal issue classification with clear progress indication.

5.3 Attorney Case Management

Attorneys work from a denser, data-forward layout suited to managing a full caseload — summary stat cards up top, followed by a filterable, sortable case table with status badges and upcoming hearing dates highlighted in Alert Terracotta.



Attorney Case List

Caseload overview with status tracking and hearing deadlines.

6. Sketch UI Sample Screenshots

The wireframes below are low-fidelity sketches representing the primary screens of the Themis platform. They illustrate layout and information hierarchy ahead of high-fidelity visual design.

5.1 Citizen Dashboard — Case Journey View

THEMIS	Case Journey Timeline	Case Summary
☐ Legal Search ☐ Guided Assessment ☐ My Documents ☐ My Case	● Assessment ○ Document Filed ○ Attorney Assigned ○ Hearing Scheduled ○ Resolved	Attorney: -- Jurisdiction: -- Next Hearing: --
	[Continue Assessment]	⚠ Reminder: Upload ID proof

5.2 Attorney Case List View

Case ID	Citizen	Status	Next Hearing
CASE-1042	Citizen 1	Filed	--
CASE-1051	Citizen 2	In Review	--
CASE-1063	Citizen 3	Scheduled	24 Aug

5.3 Document Upload & OCR Status

Upload Supporting Documents	
 Drag files here or click to browse Accepted: PDF, JPG, PNG • Max 10MB	
✓ aadhar_card.pdf	OCR: Complete
✓ rent_agreement.pdf	OCR: Processing...
✓ photo_evidence.jpg	OCR: Complete